Using Dilations to Reason About Similarity

Grade 8 / Pre-Algebra — Geometry

Image coordinates, scale factor from lengths, slope under a dilation, proportions in similar figures, and how area scales

Notation and rules used on this sheet:

- A **dilation** with centre C and scale factor k sends each point P to P' so that P' lies on ray CP with $CP' = k \cdot CP$.
- Centre at the origin: $(x, y) \longmapsto (kx, ky)$. Centre at (h, m): $(x, y) \longmapsto \big(h + k(x - h),\ m + k(y - m)\big)$.
- A dilation multiplies every length by k and leaves every angle measure unchanged, so a figure and its image are **similar**.
- Show the ratio you divided, and label every answer with its unit.

1. A logo is drawn on a coordinate grid. The corner point is $P(3, 5)$, in grid units. The design is dilated about the origin with scale factor $k = 2$. Find the coordinates of the image point P'. Give the x-coordinate and the y-coordinate separately, and say in one sentence how you got each one.

Answer: ________________

2. The same logo has a second corner at $Q(8, -6)$, in grid units. This time it is dilated about the origin with scale factor $k = 0.5$. Find the coordinates of the image point Q', and state whether the image is an enlargement or a reduction of the original.

Answer: ________________

3. On the grid, segment $\overline{AB}$ runs from $A(0,0)$ to $B(6,8)$, in grid units. After a dilation about the origin, its image $\overline{A'B'}$ runs from $A'(0,0)$ to $B'(9,12)$.

(a) Find the length AB.

Answer: ________________ units

(b) Find the length $A'B'$.

Answer: ________________ units

(c) Use the two lengths to find the scale factor k of the dilation.

Answer: ________________

4. A ramp is drawn as the segment from $(2,3)$ to $(6,9)$ on a grid. It is dilated about the origin with scale factor $k = 3$, giving the image segment from $(6,9)$ to $(18,27)$. Find the slope of the original segment and the slope of the image segment. Then state what your two answers show about the two segments.

Answer: ________________

5. Triangle ABC is dilated about the origin with scale factor $k = 3$ to give triangle $A'B'C'$. Explain, in two or three complete sentences, why $\triangle A'B'C'$ must be *similar* to $\triangle ABC$. Your explanation must say what the dilation does to the three side lengths and what it does to the three angle measures.

6. A photograph measures 4 centimetres by 6 centimetres. It is enlarged for a display so that the 6 centimetre side becomes 15 centimetres.

(a) Find the scale factor of the enlargement.

Answer: ________________

(b) Find the length of the enlarged short side, in centimetres.

Answer: ________________ cm

7. A dilation has centre $C(1,1)$ — *not* the origin — and scale factor $k = 3$. The point $M(3,2)$ is on the design. Use $(x,y) \longmapsto \big(h+k(x-h),\ m+k(y-m)\big)$ with $h = 1$ and $m = 1$ to find the image point M'. Give the x-coordinate and the y-coordinate separately.

Answer: ________________

8. A triangular sail has sides of 6 centimetres and 8 centimetres on a scale model. A larger sail is similar to the model, and the side matching the 6 centimetre side measures 9 centimetres.

(a) Find the scale factor from the model to the larger sail.

Answer: ________________

(b) Find the length of the side matching the 8 centimetre side, in centimetres.

Answer: ________________ cm

9. A rectangular tile is drawn with vertices $(0, 0)$, $(4, 0)$, $(4, 3)$ and $(0, 3)$, in grid units. It is dilated about the origin with scale factor $k = 2$, giving image vertices $(0, 0)$, $(8, 0)$, $(8, 6)$ and $(0, 6)$.

(a) Find the area of the original tile.

Answer: ________________ square units

(b) Find the area of the image tile.

Answer: ________________ square units

(c) Divide the image area by the original area. Explain how that number is related to k.

Answer: ________________

10. On a scale drawing of a school yard, a flagpole is 8 centimetres tall and the wall beside it is 6 centimetres wide. The real wall is 21 metres wide. The drawing and the real yard are similar figures. Find the scale factor from the drawing to the real yard, then find the real height of the flagpole in metres. Show the ratio you used.

Answer: ________________ m